Fiscal Year:	FY 2008	**Task Last Updated:**	FY 11/12/2008
PI Name:	Obenaus, Andre Ph.D.		
Project Title:	Non-Invasive Assessment Of Neuropathology Following CNS Radiation Exposure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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PI Email:	aobenaus@llu.edu	**Fax:**	FY 909-558-0320
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-7108
Organization Name:	Loma Linda University		
PI Address 1:	Chan Shun Pavilion, Room A-1010		
PI Address 2:	11175 CAMPUS ST		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92354	**Congressional District:**	41
Comments:			
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 6/30/2008 per P. Denkins/JSC (8/07) End date changed to 6/30/2007 per JSC update (3/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nelson, Gregory (Loma Linda University)		
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Task Description:	High energy charged particle (HZE or high LET) radiation effects on the central nervous system are poorly understood. Furthermore, the development of biomarkers for neuronal injury is desirable, particularly if these are non-invasive. Magnetic resonance imaging (MRI) and its associated modalities have been used to study disease processes with great success (i.e. cancer). However, lacking is a clear temporal and spatial description of heavy ion or proton radiation effects on brain tissue. We propose to test the hypothesis that MRI can readily identify regional alterations within the brain following radiation exposure. In addition, these non-invasive imaging modalities will correlate with inflammatory and other histological correlates of neurodegeneration. Preliminary data collected by our laboratory has shown that T2 and diffusion weighted imaging can detect neuronal changes within the brain following 2 Gy 56Fe exposure. In addition, magnetic resonance spectroscopy also revealed transient decreases in brain metabolite levels at 1 mo after radiation exposure. These findings are in agreement with previous work in our laboratory on other model systems which demonstrate that these MRI visible alterations relate to ongoing neuropathology. The significance of this proposal is that we will have elucidated the underlying neurodegenerative events (temporally and spatially) that are associated with radiation exposure using non-invasive imaging modalities. This research could allow space explorers to be monitored effectively using “biomarkers” visible with MRI. Finally, this research will provide the basis for non-invasive assessment of biological countermeasures (pharmacological) to prevent these progressive alterations in the brain. Research Emphasis: CNS Radiobiology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This grant demonstrated that non-invasive magnetic resonance imaging (MRI) can identify alterations within the brain after a single exposure to 56Fe irradiation. We used multiple imaging modalities including: magnetic resonance spectroscopy (MRS) to directly measure brain metabolites, such as choline, creatine and lactate; T2- weighted imaging (T2WI) to quantitatively evaluate brain water content that is a reflection of ongoing tissue changes; T1-weighted imaging (T1WI) with contrast agents to evaluate the integrity of the blood brain barrier (BBB); and diffusion weighted imaging (DWI) which measures water mobility within the brain and is sensitive ongoing neuropathology. Imaging was performed over the course of 2 years after a single exposure to 1, 2, or 4 Gy of 56Fe radiation. The key findings is that multi-modal MRI can detect neuronal changes within the various brain regions as early as 1 week after exposure and that these imaging changes evolve over the course of 2 years. In the early phase (0-3 mo) after radiation exposure, DWI is the most sensitive to ongoing tissue matrix modifications. From 3mo to 2 years both T2WI and DWI show differential responses that can be used as biomarkers of altered brain tissue. Many of these imaging changes were often observed without gross histological changes. At the early time points microglial responses could be observed with subsequent astrogliosis in selected animal groups. In addition, magnetic resonance spectroscopy also revealed transient decreases in brain metabolite levels as early as 1 mo after radiation exposure. The significance of these findings are that we able to elucidate the underlying tissue changes in advance of neurodegenerative events (temporally and spatially) that are associated with radiation exposure using non-invasive imaging modalities. This research could allow space explorers to be monitored effectively using “biomarkers” visible with MRI. In conclusion, this research could provide the basis for non-invasive assessment of biological countermeasures (pharmacological) to prevent these progressive alterations in the brain.
Task Progress:	This grant focused on non-invasive assessment of radiation altered physiology at 1 week through 18 mo after a single brain-only dose (1, 2, 4 Gy) of 56Fe. Correlative histochemistry and immunohistochemistry was performed to validate and assess underlying tissue matrix changes within the brain at each time point. Often there were no overt tissue abnormalities that could be visualized using conventional imaging, either by T1WI, T2WI or DWI. Thus, visual evaluation of standard MR images did not reveal any regions of abnormality. While there were rare cases of contrast enhancement in some animals after radiation – there were more animals with contrast enhancement at higher doses – these findings were not statistically significant. One of the great advantages of MRI is that there are multiple contrast levels (i.e. T1, T2, DWI and MRS) and that these contrasts can be quantified. While MR definitions are beyond the scope of this report, biophysical tissue properties such as T2 relaxation values, apparent diffusion coefficients and ratiometric metabolite quantification can be used to report tissue changes prior to overt visual changes. Thus, MRI/MRS could report tissue level changes in the brain prior to wholesale histological damage. This is in fact an overriding finding from the present studies – MRI can highlight quantitative changes in the brain after radiation in the absence of overt histopathology. Examination of the hippocampus as early as 1 week after a single dose of radiation, T2 values exhibited significant prolongations that occurred in concert with and ADC reductions. Increased T2 values are often associated with increased water content while ADC decreases are reflective of reduced water mobility. At this early time point, there was no observable neuronal pathology but immunohistochemistry for astrocytes found increased glial hypertrophy at 4 Gy animals. Thus, MRI may provide non-invasive assessment of the initial molecular/cellular disturbances in vivo after HZE irradiation prior to overt cellular changes. Similarly, at 1 month post-irradiation we found a significant increase in T2 values at 2 Gy compared to 0 Gy. However, the apparent diffusion coefficient (ADC) revealed an inverse dose-dependent quantitative change in water mobility. Compared to 0 Gy, the ADC increased 122% at 1 Gy and declined to 44% above control levels at 4 Gy. Thus, the greatest water mobility was evident at 1 but not 4 Gy. Spectroscopy analysis of hippocampal metabolites showed a significant increase in the N-acetylaspartate/choline ratio at 4 Gy and the appearance of a lactate peak. The presence of a lactate peak is consistent with neuronal damage as similar changes have been found in other neuropathological disease states. Interestingly, histology demonstrated no overt pathological changes in neuronal and astrocyte populations. However, a significant inverse dose-dependent morphological change in the microglial population was detected in irradiated animals that varied with dose. At 1 month after tissue matrix modifications induced by 56Fe particle radiation we could detect by MRI fundamental changes in the structure and function of the hippocampus. Long-term monitoring of radiation induced changes within the brain was evaluated with MRI methods for 3-18 months post-irradiation. Again, memory related structures were monitored using T2-, diffusion-weighted, pre and post-contrast enhanced T1-weighted imaging. MRI was performed at 0.25, 1, 3, 6, 9, 12 and 18 mo and T2 relaxation times and apparent diffusion coefficients (ADC) were quantified within the hippocampus (HP), entorhinal cortex (EC), retrosplenial cortex (RC) and thalamus (Thal). In the absence of visible lesions on the multi-modal MR images, ADC and T2 relaxation times revealed 56Fe irradiation induced-dynamic changes in all the ROIs over the 18 mo time course. The oscillatory pattern of increases followed by decreases in the first 3 mo was then

	followed by normalization greater than 9 months. These oscillatory patterns of MR changes were spatially similar within the each region although the magnitude varied. Of interest was that finding that both immunohistochemical labeling of astrocytes and microglia responded in an inverse fashion to the ADC results. Therefore, quantitative MRI provides a non-invasive approach to monitor temporal and spatial brain tissue modifications after 56Fe irradiation. ADC is a sensitive metric to reveal the ongoing dynamics of multiple cellular compositions in vivo. Magnetic resonance spectroscopy (MRS) was sensitive at 1 month to hippocampal changes with altered NAA/Cho combined with the appearance of al lactate peak. These findings were extended to evaluate the 3-18 months post-irradiation time points. There was a significant early drop in the NAA/PCr ratio at one week that was observed over all doses. The 4 Gy dose produced the most dramatic effects; at 6 months a doubling of the NAA/PCho ratio and at 18 months this ratio was halved. While at the early time points radiation at 4 Gy evoked MRS-observable changes, at later time points this was not the case, based in part on the lack of lactate in the spectra. Based on these results combined with histological and immunohistochemical findings we surmise that non-neuronal cells were most sensitive to the high LET radiation, where we found gliosis evolved over the 18 months. These results further strengthen our contention that MR based signatures can be useful for monitoring the physiological changes within the brain after 56Fe radiation. The significance of these findings is that we are able to elucidate the underlying tissue changes in advance of neurodegenerative events (temporally and spatially) that are associated with radiation exposure using non-invasive imaging modalities. This research could allow space explorers to be monitored effectively using “biomarkers” visible with MRI. In conclusion, this research could provide the basis for non-invasive assessment of biological countermeasures (pharmacological) to prevent these progressive alterations in the brain.
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